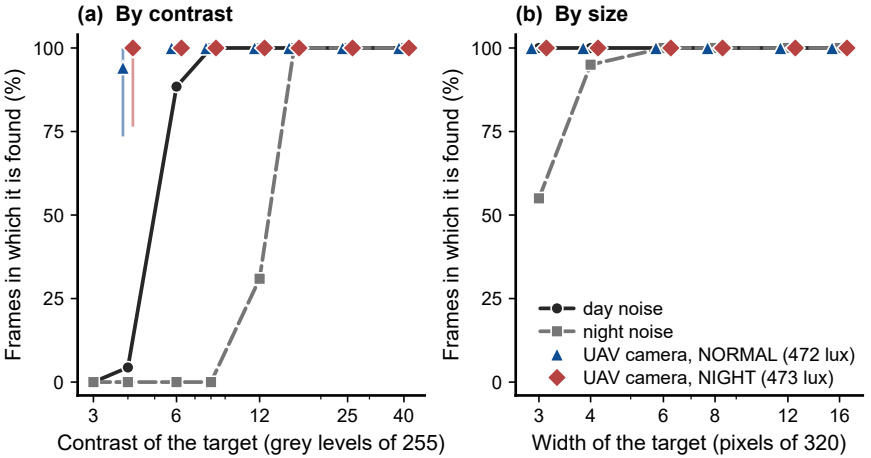
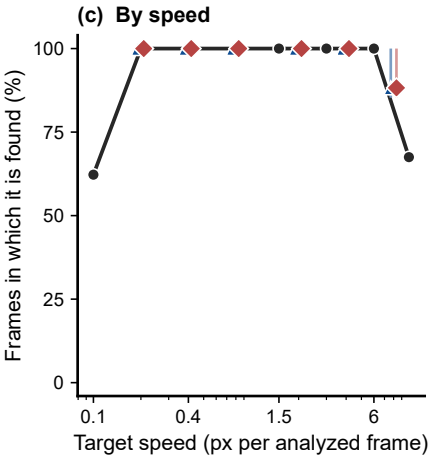
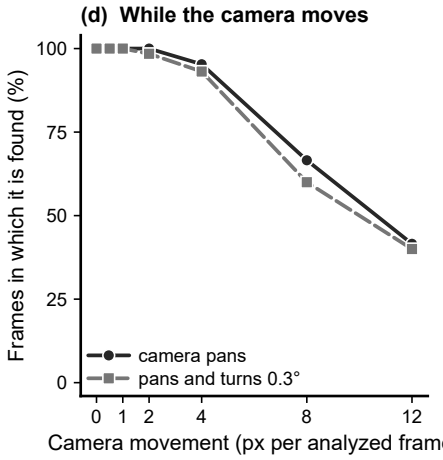






(e) Nothing moves: regions reported

Generated scene	medium	high
still, day noise	0 in 320	0 in 320
still, night noise	0 in 320	0 in 320
lamp flicker 5 %	0 in 320	0 in 320
exposure wanders 2 % a frame	0 in 320	3 in 320
pan 2.3 px/frame	0 in 320	0 in 320
pan 6.5 px + shake 0.5 px	0 in 320	0 in 320
pan + turn 0.3 deg/frame	0 in 320	2 in 320
pan + zoom 0.5 %/frame	1 in 320	5 in 320
dim flat scene, night, row noise	0 in 320	0 in 320
dim flat scene, pan 2.3	0 in 320	0 in 320

Camera pans (d), nothing moves: 1 in 2240 analyzed frames.
UAV camera, NORMAL, nothing drawn in: 13 in 102 analyzed frames*
UAV camera, NIGHT, nothing drawn in: 18 in 102 analyzed frames*
The scene left to itself for 5 min: 12 movements,
1,323 of 2,619 analyzed frames with a region*
* a lit room with people in it: what moved there,
or a false alarm; the pictures cannot be kept to tell.

(f) What the detector costs the UAV

Median of the stretches	off	on
Pi CPU, all four cores (%)	9.1	11.8
Processor clock (MHz)**	800	1800
Camera frames per second	30.0	30.0
Video frames lost	0 of 3,686	0 of 3,617
Seal and send per video record (µs)	149	94
Full handshake at the UAV (ms)	21.9	21.1
1-RTT Cached RapidRekey (ms)	9.6	8.1
PQ ratchet (ms)	15.1	10.9

One analyzed frame: 31 ms, 8 a second, 320 × 180 pixels, beside video of 1280 × 720. Off and on in turns, 4 stretches of 30 s each.
** with the detector on, the Pi's governor raises the clock:
that is why the other work gets quicker, not slower.
(a) to (c): target 8 pixels wide, contrast 25, 2 pixels per analyzed frame unless varied. Grey: generated scenes. Colour: the UAV's camera, targets drawn in on the UAV; median of 3 captures, thin line: their range.